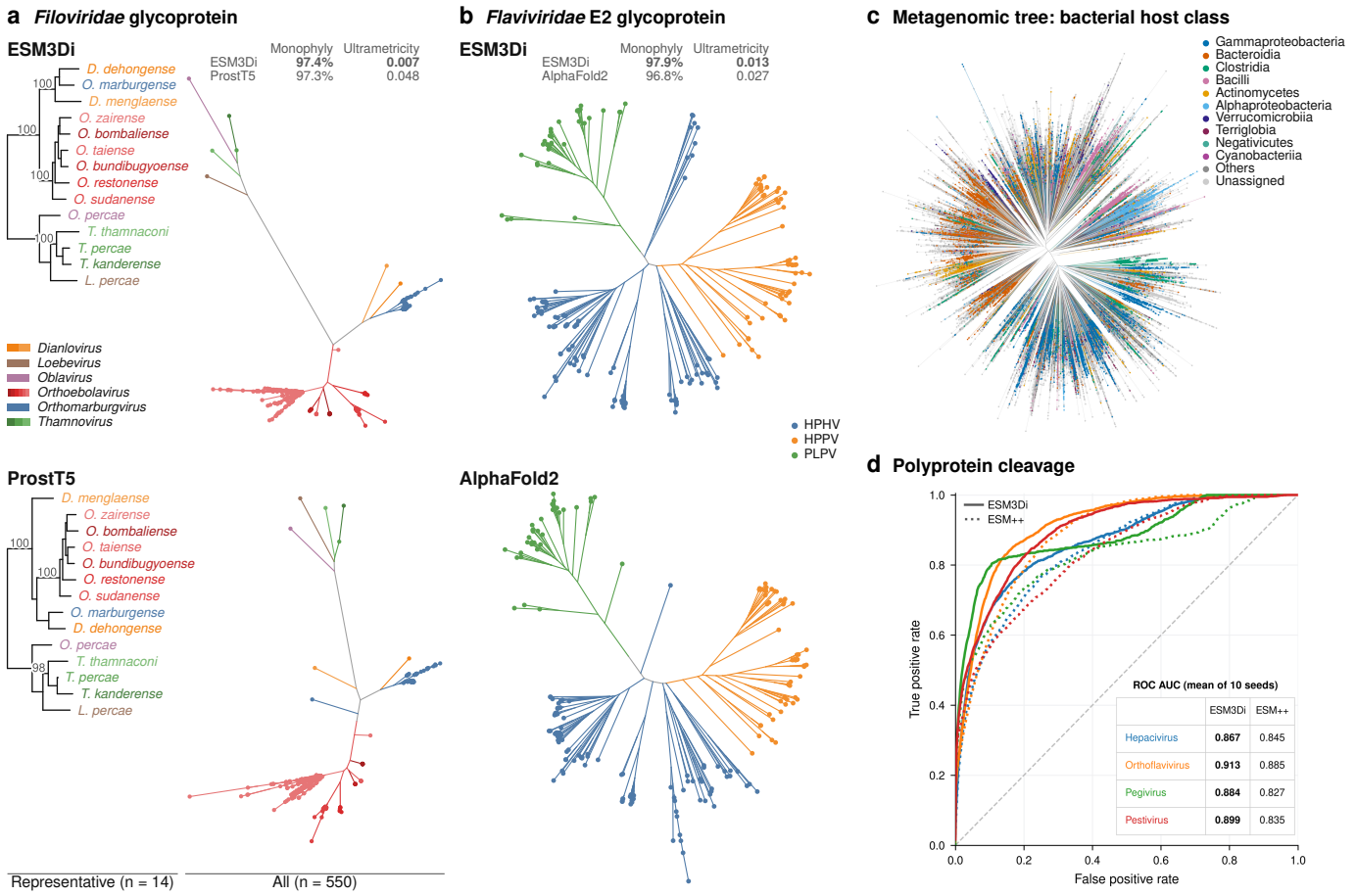

a Filoviridae glycoprotein
ESM3Di
Monophyly Ultrametricity
ESM3Di 97.4% 0.007
ProstT5 97.3% 0.048
D. dehongense
O. marburgense
D. menglaense
O. zairense
O. bombaliense
O. taiense
O. bundibugyoense
O. restonense
O. sudanense
O. percae
T. thamnaconi
T. percae
T. kanderense
L. percae
100
100
100
100
Dianlovirus
Loebevirus
Oblavirus
Orthoebolavirus
Orthomarburgvirus
Thamnovirus
ProstT5
D. menglaense
O. zairense
O. bombaliense
O. taiense
O. bundibugyoense
O. restonense
O. sudanense
O. marburgense
D. dehongense
O. percae
T. thamnaconi
T. percae
T. kanderense
L. percae
100
100
98
Representative (n = 14)
All (n = 550)
b Flaviviridae E2 glycoprotein
ESM3Di
Monophyly Ultrametricity
ESM3Di 97.9% 0.013
AlphaFold2 96.8% 0.027
HPHV
HPPV
PLPV
AlphaFold2
c Metagenomic tree: bacterial host class
Gammaproteobacteria
Bacteroidia
Clostridia
Bacilli
Actinomycetes
Alphaproteobacteria
Verrucomicrobiia
Terriglobia
Negativicutes
Cyanobacteriia
Others
Unassigned
d Polyprotein cleavage
ESM3Di
ESM++
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC AUC (mean of 10 seeds)
ESM3Di ESM++
Hepacivirus 0.867 0.845
Orthoflavivirus 0.913 0.885
Pegivirus 0.884 0.827
Pestivirus 0.899 0.835